MIL-D-43490A

25 March 1971
SUPERSEDING
MIL-D-43490
27 March 1967

MILITARY SPECIFICATION

DOUGH MIXING AND MAKE-UP OUTFIT, TRAILER-MOUNTED, M-534

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers a trailer-mounted dough-mixing and make-up outfit having all necessary equipment for the mixing, dividing, moulding and handling of dough.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-C-30	-	Cable and Wire, Electrical (Power, Fixed Installation).
J-C-580	-	Cord, Flexible, and Wire, Fixture, (Electrical 0-to 600-Volt Service).
O-E-910	-	Extinguisher, Fire, Carbon-Dioxide (Hand and Wheeled Types).
T-T-871	-	Twine, Cotton, Wrapping.
W-F-406	-	Fittings For Cable, Power, Electrical and Conduit, Metal, Flexible.

FSC 7320

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W-L-101	-	Lamp, Incandescent, (Electric, Large, Tungsten-Filament).
CC-M-636	-	Motors, Alternating-Current, (Fractional Horsepower).
CC-M-641	-	Motor, Alternating Current, (Integral Horsepower, 200 HP and Smaller).
FF-B-171	-	Bearings, Ball, Annular (General Purpose).
FF-B-185	-	Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	-	Bearing, Roller, Tapered.
FF-P-386	-	Pins, Cotter (Split).
FF-R-556	-	Rivets, Solid, Small; Rivets, Split, Small; Rivets, Tubular, Small; and Caps, Rivets; General Purpose.
HH-C-576	-	Cork Sheet (Gaskets, Sheets and Strips).
HH-G-156	-	Gasket Material, General Purpose; Rubber Sheets, Strips, and Special Shapes.
HH-P-106	-	Packing Material, Flax or Hemp.
QQ-A-225/1	-	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 1100.
QQ-A-225/8	-	Aluminum Alloy Bar, Rod, Wire, and Special Shapes, Rolled, Drawn, or Cold Finished, 6061.
QQ-A-250/1	-	Aluminum Alloy 1100, Plate and Sheet.
QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet.
QQ-A-250/11	-	Aluminum Alloy 6061, Plate and Sheet.
QQ-A-601	-	Aluminum Alloy Sand Castings.
QQ-B-626	-	Brass, Leaded and Non-Leaded; Rods, Shapes, Forgings, and Flat Products with Finished Edges (Bars, Flat Wire and Strip).
QQ-B-1005	-	Bronze Castings, Leaded Tin and High Leaded Tin.
QQ-C-390	-	Copper Alloy Castings (Including Cast Bar).
QQ-I-652	-	Iron Castings, Gray.
QQ-I-666	-	Iron, Malleable, Ferritic, For Castings.
QQ-S-624	-	Steel Bar, Alloy, Hot Rolled and Cold Finished, (General Purpose).
QQ-S-634	-	Steel, Bar, Carbon, Cold Finished (Standard Quality).
QQ-S-681	-	Steel Castings.
QQ-S-682	-	Steel, Corrosion-Resisting-Clad Plates, Sheets, and Strips.
QQ-S-741	-	Steel, Carbon: Structural Shapes, Plates, and Bars.
QQ-S-781	-	Strapping, Steel, Flat and Seals.
QQ-T-425	-	Tinplate (Hot Dip and Electrolytic).
QQ-T-830	-	Tube, Steel, (Carbon, Mechanical; Seamless and Welded).
QQ-W-423	-	Wire, Steel, Corrosion-Resisting.
QQ-W-428	-	Wire, Steel, High Carbon, Round, Uncoated for Mechanical Springs, (General Purpose).

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QQ-W-461	-	Wire, Steel, Carbon (Round, Bare and Coated).
QQ-W-470	-	Wire, Steel, Carbon, Spring, Music.
RR-C-271	-	Chains and Attachments, Welded, Weldless, and Roller Chain.
TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.
TT-P-666	-	Primer Coating, Zinc Yellow (Zinc Chromate), For Aluminum Surfaces.
WW-C-566	-	Conduit, Metal, Flexible.
WW-C-581	-	Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple Electrical Conduit: Zinc Coated.
WW-N-351	-	Nipples, Pipe, Threaded.
WW-P-406	-	Pipe, Steel (Seamless and Welded)(For Ordinary Use).
WW-P-471	-	Pipe Fittings: Bushings, Locknuts, and Plugs; Iron, Steel, and Aluminum; (Threaded): 125-150 Pounds.
WW-P-521	-	Pipe Fittings, Flange Fittings and Flanges, Steel and Malleable Iron (Threaded and Butt-Welding) 150 Pound.
MMM-A-260	-	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible.
PPP-C-650	-	Crates, Wood, Open and Covered.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging

MILITARY

MIL-P-116	-	Preservation, Methods Of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-1144	-	Pipe, Stainless Steel, (Corrosion-Resistant), Seamless or Welded.
MIL-C-2221	-	Covers, Cotton Duck, Mobile Bakery Plant.
MIL-R-3065	-	Rubber, Fabricated Products.
MIL-F-3541	-	Fittings, Lubrication.
MIL-B-5687	-	Bearings, Sleeve; Washers, Thrust, Sintered, Metal Powder, Oil-Impregnated.
MIL-S-7947	-	Steel, Sheet and Strip (1095) Aircraft Quality.
MIL-M-10578	-	Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base).
MIL-S-11183	-	Scales, Dial and Beam Indicating, Baker's, Dough Piece Checking.

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MIL-S-13281 - Steel, Carbon, Alloy and High Strength Low Alloy; For Welded Structures;
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.

STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability.
FED-STD-151 - Metal, Test Methods.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-417 - Rubber, Composition, Vulcanized, General Purpose, Solid (Symbols and Tests).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-642 - Identification marking of Administrative, Combat, and Tactical Support Vehicles.
MS 35647 - Padlock, Key Operated.

PUBLICATIONS

Department of the Army Technical Manual TM-10-410 Bread Baking

DRAWINGS

ARMY NATICK LABORATORIES

5-11-543 - Dough Mixing and Make-up Outfit, Trailer Mounted, M-534; Equipment Assembly Complete.
5-11-555 - Dough Mixer Assembly.
5-11-556 - Dough Mixer, Longitudinal and Miscellaneous Sections.
5-11-557 - Dough Mixer, Right End and Miscellaneous Sections.
5-11-558 - Dough Mixer, Frame Assembly.
5-11-559 - Dough Mixer, Frame Details.
5-11-560 - Dough Mixer, Frame and Miscellaneous Details.
5-11-561 - Dough Mixer, Tempering Assembly and Details.

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5-11-562 - Dough Mixer, Tempering Tank Details A.
5-11-563 - Dough Mixer, Tempering Tank Details B.
5-11-564 - Dough Mixer, Right Top Cover Assembly and Front Door, Assembly.
5-11-565 - Dough Mixer, Bowl Assembly and Details.
5-11-566 - Dough Mixer, Bowl Worm and Box Details.
5-11-567 - Dough Mixer, Worm Box and Miscellaneous Details.
5-11-568 - Dough Mixer, Worm Box Details and Drum Hand Wheel Assembly.
5-11-569 - Dough Mixer, Left Side Trunnion and Miscellaneous Details.
5-11-570 - Dough Mixer, Right Side Trunnion and Miscellaneous Details.
5-11-571 - Dough Mixer, Agitator Assembly and Details.
5-11-572 - Dough Mixer, Agitator Details.
5-11-573 - Dough Mixer, Bowl Cover Assembly and Instruction Plate.
5-11-574 - Dough Mixer, Drive Housing.
5-11-575 - Dough Mixer, Drive Housing Cover and Details.
5-11-576 - Dough Mixer, Drive Housing, Details A.
5-11-577 - Dough Mixer, Drive Housing, Details B.
5-11-578 - Dough Divider, Assembly.
5-11-579 - Dough Divider, Longitudinal and Miscellaneous Sections.
5-11-580 - Dough Divider, End and Miscellaneous Section.
5-11-581 - Dough Divider, Frame Assembly and Details.
5-11-582 - Dough Divider, Frame and Miscellaneous Details.
5-11-583 - Dough Divider, Right Door and Top Lever Cover.
5-11-584 - Dough Divider, Rear and Left Door.
5-11-585 - Dough Divider, Dough Box and Details.
5-11-586 - Dough Divider, Dough Box Top.
5-11-587 - Dough Divider, Crank Shaft and Details.
5-11-588 - Dough Divider, Spring Bracket and Details.
5-11-589 - Dough Divider, Knife, Dough Plunger, and Details.
5-11-590 - Dough Divider, Cylinder Details A.
5-11-591 - Dough Divider, Cylinder Details B.
5-11-592 - Dough Divider, Cylinder Details C.
5-11-593 - Dough Divider, Cylinder Details D.
5-11-594 - Dough Divider, Conveyor Details A.
5-11-595 - Dough Divider, Conveyor Details B.
5-11-596 - Dough Divider, Conveyor Details C.
5-11-597 - Dough Divider, Top Duster.
5-11-598 - Dough Divider, Lower Duster.
5-11-599 - Dough Divider, Gear Box and Details.
5-11-600 - Dough Divider, Mitre Gear Box and Details.
5-11-601 - Dough Divider, Gear Box Details.
5-11-602 - Dough Divider, Hopper Details and Instruction Plate.

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5-11-603 - Dough Divider, Oil Container and Miscellaneous Details.
5-11-621 - Electrical Panel.
5-11-622 - Wiring Diagram.
5-11-624 - Repair Part and Dough Thermometer Box.
5-11-625 - Dough Divider, Weight Box, Scale Support and Hold Down Bracket.
5-11-681 - Water Transfer System for Dough Mixer, Assembly and Details.
5-11-682 - Water Transfer System for Dough Mixer, Details.
5-11-829 - Dough Moulder, Assembly.
5-11-830 - Dough Moulder, Right Side View.
5-11-831 - Dough Moulder, Left Side View.
5-11-832 - Dough Moulder, Conveyor Assembly.
5-11-833 - Dough Moulder, Details A.
5-11-834 - Dough Moulder, Details B.
5-11-835 - Dough Moulder, Conveyor Details A.
5-11-836 - Dough Moulder, Pressure Board Positioner, Assembly and Details.
5-11-837 - Dough Moulder, Conveyor Details B.
5-11-838 - Dough Moulder, Pressure Plate Assembly.
5-11-839 - Dough Moulder, Assemblies and Details.
5-11-840 - Dough Moulder, Details C.
5-11-841 - Dough Moulder, Right Head Frame.
5-11-842 - Dough Moulder, Left Head Frame.
5-11-843 - Dough Moulder, Details D.
5-11-844 - Dough Moulder, Motor Gear and Idler Details.
5-11-845 - Dough Moulder, Rear Scraper and Roll Details.
5-11-846 - Dough Moulder, Roll Details.
5-11-847 - Dough Moulder, Front Scraper Support and Details.
5-11-848 - Dough Moulder, Duster Box Details A.
5-11-849 - Dough Moulder, Duster Box Details B.
5-11-850 - Dough Moulder, Details F.
5-11-851 - Monorail System, Assembly and Sections.
5-11-852 - Monorail System, Sections and Details.
5-11-853 - Monorail System, Details A.
5-11-854 - Monorail System, Details B.
5-11-855 - Lighting Power Systems, Assembly and Details.
5-13-2176 - Lighting Kit, Mobile Bakery.

(Miniature copy of Drawing 5-11-543, identified as figure 1 is attached for information purposes only.)

(Copies of specifications, standards, drawings, and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

Standard No. 73 - Motor-Operated Appliances
Standard No. 489 - Circuit-Breakers, Branch Circuit and Service
Standard No. 508 - Industrial Control Equipment

(Application for copies should be addressed to the Underwriter's Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1284 Walt Whitman Road, Melville, L.I., New York 11749; and 1655 Scott Boulevard, Santa Clara, California 95050.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Standard E18-67 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
Standard A296-62T - Corrosion-Resistant Iron-Chromium and Iron-Chromium Nickel Alloy Castings for General Application
Standard D2016-65 - Moisture Content of Wood

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Standard B29-1 - Transmission Roller Chains and Sprocket Teeth (SP-69)

Standard B29-2 - Inverted Tooth (Silent) Chains and Sprocket Teeth (SP-68)

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, N.Y. 10008.)

INTERSTATE COMMERCE COMMISSION

Motor Carrier Safety Regulations

(Application for copies should be addressed to the Interstate Commerce Commission, Bureau of Motor Carriers, Washington, D. C. 20423 or the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Department of Defense Materials on Open Top Cars

Phamphlet MD 7 - Rules Governing the Loading of Commodities on Open Top Cars

(Application for Copies should be addressed to the Associated Railroads, 59 East Van Buren Street, Chicago, Illinois 60605.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Alternate components.- Components offered as equivalent to components specified hereinafter and on the referenced drawings as a specific manufacturer's part number or equal shall be functionally equal, or equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the preproduction sample, the supplier shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of preproduction sample manufacture and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2 and 6.7).

* 3.3 Safety standards.- All electrical equipment and material shall conform to UL Standards 508, 489, and 73 as applicable.

* 3.3.1 Compliance.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the electrical components and materials of the dough mixing and make-up outfit he proposes to furnish under this specification meet the applicable requirements of UL Standards 508, 489 and 73. Acceptable evidence of meeting the applicable requirements of the UL shall be the UL label or listing mark or a

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certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the electrical components and materials of the dough mixing and make-up outfits offered have been tested and conform to the applicable requirements of UL Standards 508, 489 and 73.

* 3.4 Materials.- The materials shall conform to the applicable specifications and requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used by the manufacturer for dough mixing and make-up outfits provided the completed end item complies with all provisions of this specification.

3.4.1 Steel, carbon.-

3.4.1.1 Sheet or strip.- The carbon steel sheet or strip specified on the drawing shall conform to the chemical composition of AISI steel numbers 1010 through 1020 of FED-STD-66.

3.4.1.2 Bar.- Steel bar shall conform to QQ-S-634.

3.4.1.3 Angles and channels (structural size).- Structural size steel angles channels, and the monorail tee bars shall conform to QQ-S-741.

3.4.1.4 Angles (bar size).- Bar size steel angles shall conform to MIL-S-13281.

3.4.1.5 Tube.- Steel tube shall conform to QQ-T-830.

3.4.1.6 Castings.- Steel castings shall conform to QQ-S-681. The dough divider levers and collars shall be fabricated of class 65-35 material and the moulder agitator ends shall be fabricated of class 80-50 material.

3.4.1.7 Spring steel strip.- Spring steel strip shall conform to MIL-S-7947.

3.4.1.8 Pipe.- Steel pipe shall conform to WW-P-406. The pipe used for nipples, coupling, and elbows, shall be galvanized pipe conforming to weight A, class 1 and the pipe used for coupling the dough divider pressure fittings to the oil lines shall be class 2.

3.4.1.9 Pipe plug.- The 1/4 inch, square head, steel pipe plug used in the dough divider gear box shall conform to type I of WW-P-471.

3.4.1.10 Nipples, steel or wrought iron.- Steel or wrought iron nipples shall conform to type I of WW-N-351.

3.4.1.11 Wire.-

3.4.1.11.1 Carbon steel.- Carbon steel wire shall conform to QQ-W-461.

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3.4.1.11.2 Music.- Steel music wire used for small springs shall conform to QQ-W-470.

3.4.1.11.3 Spring steel.- Spring steel wire used for the divider discharge lever shaft springs shall conform to type II of QQ-W-428. Wire used for the monorail latching bar springs and swing link springs shall conform to type I of QQ-W-428.

3.4.1.12 Alloy (4150).- The 4150 alloy steel used for the divider worm shaft shall conform to QQ-S-624.

3.4.2 Castings, gray iron.- Gray iron castings used for the dough divider and moulder shall be class 25, or better, of QQ-I-652.

3.4.3 Castings, malleable iron.- Malleable iron castings shall conform to grade II of QQ-I-666.

3.4.4 Pipe fittings, malleable iron.- Malleable iron pipe fittings shall conform to type I of WW-P-521.

3.4.5 Screen, tin plated.- Screen used for the moulder duster box shall be steel, tin plated conforming to type I, grade 1 of QQ-T-425.

3.4.6 Corrosion-resisting steel.-

3.4.6.1 Sheet or strip.- Corrosion-resisting steel sheet or strip shall conform to FED-STD-66. Unless otherwise indicated on the drawing any of the 300 classes are acceptable.

3.4.6.2 Bars.- Corrosion-resisting steel bars shall conform to FED-STD-66. Unless otherwise indicated on the drawings, any of the 300 classes is acceptable.

3.4.6.3 Wire.- Corrosion-resisting steel wire shall conform to composition 430, annealed, of QQ-W-423.

3.4.6.4 Pipe.- Corrosion-resisting steel pipe and pipe nipples shall conform to class 3 of MIL-P-1144.

3.4.6.5 Pipe fittings.- Corrosion-resisting steel pipe fittings shall be the manufacturers' standard item as used for this type of application.

3.4.6.6 Castings.- Corrosion-resistant castings used for the mixer agitator elbow, sides, caps, and bowl rib shall conform to grade CA-15 specified in ASTM A296-62T when tested as specified in 4.3.1.1. Corrosion-resisting steel castings used for the moulder rollers shall be hardenable steel and conform to either of the following two chemical compositions when tested as specified in 4.3.1.1:

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Carbon	1.55 to 1.65%	1.40% to 1.50%
Manganese	.25 to .45%	.25% to .45%
Sulphur	Less than .03%	Less than .03%
Phosphorus	Less than .03%	Less than .03%
Silicon	.20 to .35%	.20% to .35%
Chromium	12.50 to 14.00%	14% to 15%
Vanadium	.30 to .45%	.30% to .40%
Molybedenum	.70 to .90%	.30% to .40%
Cobalt	.70 to .90%	None
Iron	Remainder	Remainder

3.4.6.7 Plate or sheet, clad.- Corrosion-resisting steel clad plate or sheet shall conform to FS 304, grade A of QQ-S-682.

3.4.7 Aluminum alloy.-

3.4.7.1 Sheet and strip (1100).- Aluminum alloy 1100 sheet and strip shall conform to temper H14 of QQ-A-250/1.

3.4.7.2 Plate and sheet (6061).- Aluminum alloy 6061 plate and sheet shall conform to temper T6 of QQ-A-250/11.

3.4.7.3 Plates (2024).- Aluminum alloy 2024 plates used for the moulder cover plates shall conform to temper T3 of QQ-A-250/4.

3.4.7.4 Bar stock (1100).- Aluminum alloy 1100 bar stock used for divider door thimbles shall conform to temper H14 of QQ-A-225/1.

3.4.7.5 Castings.- Aluminum castings used for the mixer worm box, drive housing, divider, and moulder parts shall conform to alloy 195 of QQ-A-601. All other aluminum castings shall conform to alloy 43.

3.4.7.6 Bars, rods, and special shapes (6061).- Aluminum alloy 6061 bars, rods and special shapes shall conform to QQ-A-225/8 and be of the temper specified on the drawings.

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3.4.8 Copper alloy.-

* 3.4.8.1 Rod.- Copper alloy rod shall conform to half hard temper, composition 360, of QQ-B-626.

* 3.4.8.2 Castings.- Copper alloy castings used for the mixer water gage and water gage pointer shall conform to alloy number B of QQ-C-390. Castings for the divider oil pump levers and the moulder eccentric caps shall conform to class C of QQ-C-390.

3.4.9 Bronze.-

3.4.9.1 Castings.- Bronze castings used for the various mixer parts shall conform to composition 8 of QQ-B-1005. Castings used for various divider-parts and moulder bearings and housing shall conform to composition 5 of QQ-B-1005.

3.4.9.2 Bearings, oil impregnated.- Oil-impregnated bronze bearing bushings shall conform to type I of MIL-B-5687.

3.4.10 Chain and attachments.-

3.4.10.1 Plumber's.- The plumber's chain attached to the divider top lever cover shall be steel, type II, class 3 or 4 of RR-C-271.

3.4.10.2 Slings.- The chain, hooks, and rings used for the dough trough carrier sling shall be steel conforming to RR-C-271. The chain shall be type I, grade C, class 1, 1/4-inch with 7-3/4-links per foot and shall be 19 links in length. The hook shall be type V, class 1 for use with 1/4-inch ring. The ring shall be type VI, 1/4-inch diameter wire by 1-1/2 inch inside diameter.

3.4.11 Electrical.-

3.4.11.1 Motors.- Electrical motors of the sizes and designs shown on the drawings shall conform to CC-M-636 or CC-M-641.

* 3.4.11.2 Cable and wire, rubber-insulated.- The rubber-insulated cable and wire for the divider, moulder, and panel shall conform to type TW, of J-C-30. The wire connected to the main bus shall be No. 4 American Wire Gage (A.W.G.) while that which is connected between the individual circuit-breakers and the mixer shall be multiple conductor wire with non-metallic sheath conforming to J-C-580.

* 3.4.11.3 Cable, armored.- Armored cable shall be lead sheathed conforming to type ACT of J-C-30.

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3.4.11.4 Conduit.- Electrical conduit shall be flexible steel conforming to WW-C-566, or rigid steel conforming to WW-C-581.

3.4.11.5 Fittings.- Cable and conduit fittings shall conform to W-F-406.

3.4.12 Conveyor belting.-

3.4.12.1 Cotton.- The cotton conveyor belt used for discharge of dough pieces from the divider shall be solid woven not less than 2-ply construction, 8 inches wide by 80 inches long, with polychloroprene latex coating with white pigment. Both ends shall have suitable and properly sized metal connecting lacing.

3.4.12.2 Duck.- The conveyor belt used for the moulding of dough pieces on the moulder shall be endless white polychloroprene impregnated, 5 plies, 9 ounce duck, one ply hard silver duck, bare top, friction surface cover, 14-3/4 inches wide by 121 inches long.

3.4.12.3 Metal.- The metal belting shall be balanced belt woven from type 430 corrosion-resisting steel wire specified in 3.4.6.3. The belts shall have a mesh of four to the inch and shall be 1/4 inch thick by 13 inches wide by 18 inches long.

3.4.13 Lumber.- Lumber used for the dough thermometer box and weight box shall be white pine, well seasoned and uniformly kiln-dried. The moisture content of the wood shall be determined in accordance with 4.3.3 and at time of fabrication shall not exceed 12 percent. The wood shall be free from decay, checks, holes, knots, shakes, warpage, splits or pitch pockets.

3.4.14 Bearings.-

3.4.14.1 Ball.- The ball bearings in the size and type specified on the drawings shall conform to FF-B-171.

3.4.14.2 Roller.- The roller bearings in the sizes and types specified on the drawings shall conform to FF-B-185 or FF-B-187, as applicable.

3.4.15 Fittings, lubrication.- Pressure lubrication fittings shall conform to MIL-F-3541.

3.4.16 Cork.- Cork sheet shall conform to type IB class 2, or type IIB class 2 of HH-C-576.

3.4.17 Packing flax.- Flax packing material shall conform to HH-P-106.

3.4.18 Chain.-

3.4.18.1 Roller.- The roller chain shall be a riveted type with a roller width of 1/2-inch and connecting links conforming to ANSI Standard B29.1.

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3.4.18.2 Silent.- The silent chain shall conform to ANSI Standard B29.2.

3.4.19 Pins, Cotter.- Cotter pins shall conform to type B, zinc-coated, or type C of FF-P-386.

3.4.20 Rivets, copper.- Copper rivets shall conform to type IV, size 1/8-inch diameter of FF-R-556 and shall be 1/2-inch long.

3.4.21 Gaskets.- The synthetic rubber used for the mixer gaskets shall be the manufacturer's standard material as used for this type of application. The rubber used for gaskets on the electrical panel shall conform to HH-G-156. The rubber for the rib gasket and water tank cover shall conform to SC 615Z of MIL-STD-417 and MIL-R-3065.

3.4.22 Padlock.- A padlock conforming to MS 35647-8 shall be furnished with the repair parts box and with the lighting kit type.

3.4.23 Bolts, screws, nuts, and washers.- Bolts, screws, nuts, washers, and other fastening devices of the size and design shown on the drawings shall be carbon steel or corrosion-resisting steel and the manufacturer's standard items. All fastening devices which could come in contact with dough shall be corrosion-resisting steel. The carbon steel bolts, screws, nuts, and washers used for the electrical panel and to secure components to the trailer deck shall be zinc or cadmium plated.

3.4.24 Instruments.- Gages, thermometers, indicators, and other instruments of the design and sizes specified on the drawings shall be the manufacturer's standard item as used for this type of application.

3.4.25 Covers, canvas.- Canvas covers shall conform to MIL-C-2221. The type I cover shall be used for the mixer, type II for the divider and moulder, and type IV for the divider hopper.

3.4.26 Scale.- The scale shall conform to class 1 of MIL-S-11183.

3.4.27 Lamp, incandescent.- Lamp incandescent shall be 100 watt, bulb A23 of W-L-101.

3.4.28 Twine, cotton.- Twine cotton shall conform to type I, 12 ply, of T-T-871.

3.4.29 Finish materials.-

3.4.29.1 Metal primer.- Primer for metal shall conform to TT-P-666.

3.4.29.2 Enamel.- Enamel shall conform to TT-E-529. Color shall be olive drab, No. X24087 conforming to FED-STD-595. For Marine Corps procurement (see 6.2), the color shall be Marine Corps green semigloss color No. 23 (see 6.3).

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3.4.29.3 Metal conditioner for aluminum and ferrous metal.- Metal conditioner for aluminum and ferrous metal shall conform to MIL-M-10578.

3.4.29.4 Primer.- Primer for the lighting kit shall conform to TT-P-636.

* 3.4.30 Fire extinguisher.- A fire extinguisher shall be furnished conforming to the requirements for type I, size 5 of O-E-910. The extinguisher shall be equipped with a quick release type bracket of the vehicle-type, having a base for mounting the extinguisher in an upright position on the trailer platform.

3.5 Design and construction.- The design and construction of the dough mixing and make-up outfit and component parts thereof shall conform to the details, dimensions, and materials shown on the drawings and as specified herein. All working parts shall fit properly and their operation shall be smooth. The assembled make-up outfit shall be tested for operation as specified in 4.4.2, for proper adjustment and interrelationship of components. During the test all lights, switches and controls shall operate as intended. The motor current shall not exceed the data plate rating.

3.5.1 Component parts.- Each dough mixing and make-up outfit shall consist primarily of a trailer, mixer, divider, moulder, monorail, electrical panel, scale, weight box, thermometers, repair part and dough thermometer box, canvas covers for mixer, divider and moulder and hopper and a lighting kit.

* 3.5.2 Electromagnetic compatibility.- Equipment procured under this specification shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.5).

3.5.3 Mixer.-

3.5.3.1 Motor, wiring and controls for the mixer.-

3.5.3.1.1 Motor and wiring for mixer.- The motor shall be as shown on Drawing 5-11-556 and as specified in 3.4.11.1 and shall be wired to the controls and power panel in conformance with the safety requirements specified in 3.3.

3.5.3.1.2 Controls for mixer.- Start, stop, jog, reset, and limit switches shall be mounted in the locations shown on applicable drawing. Start, stop, and jog push-buttons shall be assembled in the order of arrangement shown on Drawing 5-11-557, inclosed in the push-button case detailed on Drawing 5-11-560 and located as shown on Drawing 5-11-558. The distance between the start and jog push-buttons shall be such that the operator cannot engage both with one hand, to prevent starting the agitator while one hand is in the mixing bowl. A limit

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switch shall be located, wired, and adjusted to prevent starting the agitator without depressing the jog push-button simultaneously with the start push-button when the bowl is in a tilted position. A reset starter switch shall be wired and adjusted to disconnect or "cut-out" the motor under overload or low voltage.

3.5.4 Divider.-

3.5.4.1 Motor, wiring and controls for the divider.-

3.5.4.1.1 Motor and wiring for divider.- The motor shall be specified on Drawing 5-11-579 and 3.4.11.1 and shall be mounted on a base which shall be bolted to the divider base. The divider shall be completely wired, and equipped with controllers conforming to UL standards specified in 3.3. The wiring shall be installed in flexible metal conduit specified in 3.4.11.4 or shall be rubber covered cable conforming to 3.4.11.2.

3.5.4.1.2 Controls for divider.- Two manual starters shall be mounted as shown on Drawing 5-11-578 with one connected to the divider motor and the other for connection to the moulder motor. The manual starters shall be 1-1/2 horsepower(HP), 100 volt or 2 HP, 230/600 volt and shall be equipped with a 7.3-ampere heater element.

3.5.4.2 Conveyor belt.- The conveyor belt shall be as specified in 3.4.12.1 and shall be mounted below the dividing head so as to transport the pieces of dough toward the moulder rolls.

3.5.4.3 Dusters for divider.- Each divider shall be provided with a duster located as shown on Drawing 5-11-578. The duster shall be complete with adjustments for the delivery and distribution of flour and with suitable guides to direct the flour to the proper location and the conveyor belt.

3.5.5 Moulder.-

3.5.5.1 Motor and wiring for the moulder.- The motor shall be as shown on Drawing 5-11-831 and as specified in 3.4.11.1. The motor shall be mounted on the moulder frame and shall be completely protected against flour dust. The moulder shall be completely wired and enough wire and conduit for connections to the divider shall be furnished with each moulder.

3.5.5.2 Rollers.- Rollers shall be cast from corrosion-resisting steel specified in 3.4.6.6, machined, and hardened to the hardness of 62 to 65 on the Rockwell C scale when tested as specified in 4.4.4, and shall be mounted with independent adjustments for dough sheet thickness. Scrapers shall be provided and mounted on adjustable supports to insure proper scraping action on the rolls without undue heating or excessive wear.

3.5.5.3 Curling device.- The curling device shall be mounted above the conveyor belt and shall consist of a stationary metal belt riveted to the curling clamp with copper rivets. Copper rivets shall be as specified in 3.4.20.

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3.5.5.4 Duster for moulder.- Each moulder shall be provided with a duster, complete with adjustments for the delivery and distribution of flour and with suitable guides to direct the flour to the proper location at the heat rollers. A quick acting control shall be provided to permit the duster to be stopped when dough is not passing through the machine.

3.5.6 Monorail.-

* 3.5.6.1 Frame.- The monorail frame shall be fabricated from steel selected from those specified in 3.4.1 and as shown on Drawing 5-11-851 through 5-11-854. The cross tie bars shall be fabricated from 2 by 1/2 by 1/8 inch steel channel specified in 3.4.1.3. The monorail system shall be supported by four support posts shown on 5-11-852. The support posts shall fit into the stanchions which are furnished with the trailer.

3.5.6.2 Carrier rail.- The carrier rail shall be constructed of structural steel specified in 3.4.1.3 and shall be supported on the frame by cast iron carrier wheels which will permit easy rolling of one portion of the tramway to the right and one portion to the left. Provisions, as shown on Drawing 5-11-851 shall be made for filling the space created at each end when the tramway is open with a piece of rail which swings into position and forms a continuous track. Locks or stops shall be provided to control the "in" and "out" movement of the tramway. Latches shall also be provided for securing the tramway in either the open or closed position. Carriage wheels shall be provided with hydraulic pressure grease fittings for lubrication.

3.5.6.3 Carriers.- Each monorail system shall be furnished with five dough trough carriers. Each carrier shall consist of two wheel assemblies, with swivel-type yokes to permit the wheels to freely follow the rail curve contour. The yokes shall be as shown on Drawing 5-11-852 and shall support a load bar to which is welded the member with sling chains attached. Slings, hooks and chains shall be as specified in 3.4.10.2. The carrier wheels shall be equipped with precision-type ball bearings as specified in 3.4.14.1, lubricated for life and sealed against dust and dirt.

3.5.7 Panel, electrical.-

3.5.7.1 Panelboard.- The panelboard shall be as shown on Drawing 5-11-621 and shall be housed in a single cabinet consisting of two panel units of automatic circuit-breakers, fed by a central bus system. It shall provide adequate protection in the distrubution of 120-volt, single phase, lighting service and 208-volt, single phase, and three phase, power service. The panelboard shall be of the deadfront safety type constructed and arranged for assembled-unit mounting in the cabinet, so that the operator has access to the operating handles of circuit-breakers only. The circuit-breakers, shall be mounted upon steel supports and shall be removable from the front of the panelboard for inspection or replacement by removing only the cabinet front without removingthe panelboard from the

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cabinet or disturbing adjacent units. After complete assembly and attachment of the circuit-breakers to the steel supports at the back of the panel, the front shall be inclosed by a metal trim as shown on Drawing 5-11-621. In mounting the assembled units in the cabinet means shall be provided for adjustment of the unit so that the metal trim shall be in contact with the cover.

3.5.7.2 Cabinet.- The cabinet shall be of continuous welded construction and shall be fabricated from sheet conforming to 3.4.1.1. The front of the cabinet shall be inclosed by means of a cover attached to the body with steel cap screws. A rubber gasket of the material specified in 3.4.21, shall be inserted between the cover and the body and shall be secured to the body with good quality water-proof cement. A safety door shall be provided to give access to the panel units and it shall be hinged on the right side when facing the panelboard and shall be equipped with a latching device which can be easily operated with one or more handles. The latches shall be positive in action, latching at, or near, the top and the bottom of the cover, and shall be a type which will not become unlatched by vibration.

3.5.7.3 Circuit-breakers.- The circuit-breakers used on the panelboard shall conform to 3.4.11.2 and shall be capable of manual operation with an easily identified "on" and "off" position. The operating mechanism of circuit-breakers shall be such that the rupture contacts shall be quick-made and quick-break type for both manual and automatic operation. Circuit-breakers shall be rated and calibrated in accordance with the amperage and voltage specified on Drawing 5-11-621 and suitable for 60-hertz operation. The circuit-breakers shall be constructed to carry 100 percent ampere rating continuously, and trip at 125 percent ampere ratings. After calibrating, the trip element together with the operating and release mechanism, shall be sealed at the factory.

3.5.8 Covers, canvas.- Each dough mixing and make-up outfit shall be provided with canvas covers for the mixer, divider, and moulder. The covers shall be as specified in 3.4.25.

3.5.9 Box, repair parts and dough thermometer.- A repair parts box containing a dough thermometer box shall be provided and shall conform to the design, shape, dimensions, and materials specified on Drawing 5-11-624. Finishes shall be as specified in 3.6.3.

3.5.10 Box, scale, weight.- A scale weight box shall be provided and shall conform to design, shape, materials, and in the location shown on Drawing 5-11-625.

3.5.11 Lighting kit.- The lighting kit to illuminate the working area shall be as specified on Drawing 5-13-2176.

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3.5.12 Welding.- The surface of all parts to be welded shall be clean and free from scale, paint, grease, and other foreign matter. When aluminum, corrosion-resisting steel, or other ferrous metal alloy is welded, welding rod of chemical composition similar to the parts being welded shall be used. Prior to assembly, welding the components specified in 3.5.1 to the trailer chassis, all parts to be welded shall have the enamel stripped exposing bare metal for a distance of 1-inch or more on all sides of the area to be welded. Welds shall be uniformly run without voids. All welding scale shall be removed after welding has been accomplished and all projections and rough edges shall be ground smooth. Spot welds, when employed, shall be sound, smooth, and free from pits and flashes.

3.5.13 Soldering.- All surfaces to be soldered shall be thoroughly clean prior to soldering.

3.5.14 Riveting.- Rivets shall be tight. Driven heads shall be smooth, concentric with holes, and in full contact with the surface of the member riveted.

3.5.15 Heat treatment.- Sprocket teeth of the divider, and moulder and other parts specified on the drawings requiring heat treatment shall show a hardness between 50 and 55 when tested on the Rockwell C scale hardness of 52-60. The case depth of all parts requiring case-hardening shall be not less than 0.025 inch. Testing shall be as specified in 4.4.4.

3.5.16 Shearing of metal.- Shearing and chipping on all metal parts shall be done cleanly and accurately and burrs, slivers, splinters, and sharp edges shall be removed on all portions of the work to assure proper functioning of the unit when fully assembled.

3.5.17 Lubrication.- Lubrication fittings as specified in 3.4.15 shall be installed at bearings and other parts requiring pressure lubrication which do not come in contact with the dough, and shall be located for easy accessibility with a grease gun. Angle fittings shall be provided in places difficult to reach. Lubrication of all parts of the divider which come in contact with the dough shall be provided with a separate automatic lubricating system complete with a suitable oil pump, designed for the use of mineral oil. A suitable oil gage shall be provided in the location shown on Drawing 5-11-580. The oil container and cover shall be as shown on Drawi gs 5-11-603. Where necessary, oil lines shall be fabricated from pipe specified in 3.4.1.8. The bearings of the mixer shall be lubricated with a non toxic grease (see 6.6).

3.6 Finish.- All exposed metal surfaces not normally in contact with the dough shall be thoroughly cleaned of grease, scale, or other foreign substances. Metal surfaces coming in contact with the dough shall be finished to a number 2B or better finish.

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3.6.1 Aluminum and ferrous metal surface preparation.- Aluminum and ferrous metal surfaces, except the monorail wheel treads and rail treads, shall be cleaned and prepared for finishing using the compound specified in 3.4.29.3 and then given one coat of the primer specified in 3.4.29.1.

3.6.2 Enameling.- When the primer is thoroughly dried it shall be followed by one coat of enamel specified in 3.4.29.2, having a minimum dry film thickness of 0.8 mil. The electrical panel shall be given a second coat of enamel at time of assembly on the machinery trailer. The monorail surfaces requiring enamel shall have two coats. The enamel coat shall level out to produce a smooth surface of uniform color without runs, wrinkles, grits, chippings or areas of thin or no film. The enamel shall be dry and hard to the touch prior to assembly of components.

3.6.3 Repair parts and dough thermometer box.- The entire inside and outside of the repair parts box, except the outside top, shall be given a primer and enamel finish as specified in 3.6.1 and 3.6.2. The top outside shall be given a non-skid covering as shown on Drawing 5-11-624. The dough thermometer box shall be finished with two coats of clear or orange shellac of good commercial quality.

3.6.4 Duster screen.- The divider duster screen, after fabrication, shall be thoroughly cleaned of grease, scale or other foreign substance and then hot-tin dipped as specified in 3.4.5. The coating shall be smooth and free from blisters or uncoated spots and shall not plug or seal the opening in the screen.

3.7 Marking.-

3.7.1 Motors.- Each motor shall bear a securely fastened corrosion-resistant metal nameplate containing the motor manufacturer's name, manufacturer's model number, phase, cycles, voltage, rated horsepower, and allowable temperature rise.

3.7.2 Identification.-

3.7.2.1 Mixer, divider, moulder and repair parts box.- The mixer, divider, and moulder shall have the letters "U.S.", the nomenclature, manufacturer's name, contract number and date, specification number, and the stock number indicated on a photo-foil corrosion-resistant metal instruction plate in the location and size shown on Drawings 5-11-573, 5-11-602, and 5-11-845, respectively. The repair parts box shall be marked as shown on Drawing 5-11-624.

3.7.2.2 Monorail.- The letters "U.S." in characters 1/2-inch high and the nomenclature, manufacturer's name, contract number and date, specification number, and stock number in characters 1/8-inch high shall be indented, embossed or photographed on a corrosion-resistant metal nameplate and securely attached to the monorail system in a conspicuous place.

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3.7.2.3 Panelboard.- A corrosion-resistant metal nameplate bearing the manufacturer's name or trade mark, ampere capacity, voltage, number of phases, and year of manufacture shall be securely attached in a conspicuous place on the panelboard. When tested as specified in 4.4.2, the controls shall operate as intended.

3.7.2.4 Wiring.- The wiring diagram shown on Drawing 5-11-622 shall be indented, embossed, or photographed on an 11-1/2 inches by 7-7/16 inches corrosion-resistant metal plate and shall be permanently fastened to the inside surface of the panelboard door.

3.7.2.5 Circuit-breakers.- Securely fastened to the inside surface of the panelboard door shall be a metal identification card holder provided with a heavy paper card. The card shall show the intended use of each circuit in a numerical order to agree with the numbered circuit-breakers. In addition, the appropriate ampere rating shall be clearly shown on the front of each circuit-breaker so that it may be easily read upon opening the cabinet door and the manufacturer's name or trademark, electrical rating in amperes and volts, and the catalog number, or its equivalent, shall be indicated in a less conspicuous place.

3.7.2.6 Receptacles.- The voltage, phase, and intended use of each receptacle shall be indicated by a decal placed on the inside of the panelboard adjacent to the corresponding outlet.

3.7.3 Operating and lubricating instructions.- Operating and lubricating instructions shall be indicated on a photo-foil nameplate specified in 3.7.2.1, and the letters shall be of the sizes shown on the applicable drawings.

3.7.3.1 Mixer.- The following instructions shall be incorporated on the plate shown on Drawing 5-11-573:

OPERATING INSTRUCTIONS

Before starting the dough mixer, make sure bowl is thoroughly clean. Bowl can be lowered by turning hand wheel on left side of mixer. Top cover can be raised by opening clamps at sides of cover. To put water in bowl, lower bowl slightly and swing pipe from tank into bowl. Motor control is arranged so that bowl must be closed in order to mix. When bowl is open only slightly, power will be shut off, and agitator will not run unless both "jog" button and "start" button are held in at the same time. Maximum capacity of mixer is 500 pounds. Do not overload. If mixer is overloaded or voltage in power line drops, motor will automatically stop. After cause of failure is removed, it is necessary to reset relay. To reset, open bottom door and push "reset" button on starters.

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LUBRICATING INSTRUCTIONS

Clean and grease motor every 6 months. Too much grease will cause motor bearings to overheat. Motor drive chain is inclosed and rubs in a bath of oil. Check oil level in window gage on drive housing and keep level up to center of gage. Drain oil from drive housing every 6 months and refill with SAE 40 oil. Brush a small amount of SAE 40 oil on sides of roller chain every 2 weeks. This can be reached from rear of mixer. Once a week lubricate with a good grade of pressure gun grease the 7 grease fittings located as follows:

5 on left panel of mixer: 1 each for agitator bearing, pillow block, worm box, hand wheel bearing, and agitator bushing.

2 on right panel of mixer: 1 each for agitator bearing and agitator housing.

Lubricate the fitting for the motor under the front door every 2 weeks.

3.7.3.2 Divider.- The following instructions shall be incorporated on the plate shown on Drawing 5-11-602.

OPERATING INSTRUCTIONS

Before starting divider be sure dough chamber, dough plunger, knife and measuring pistons are thoroughly cleaned and amply lubricated with divider oil. Make certain dough plunger, knife, and pistons are properly assembled. Before putting dough in hopper, wipe inside of hopper with a lint free cloth saturated with divider oil. Use mineral oil, light, NF (National Formulary). Keep reservoir well filled with divider oil at all times. Keep all doors closed and all metal covers in place when divider is running. Set weight adjusting knob for desired scaling weight. Turning knob clockwise decreases weight, counterclockwise increases weight. Scaling range is 30 to 40 ounces. Do not attempt to scale dough below or above this range.

LUBRICATING INSTRUCTIONS

Motor ball-bearings are pre-packed and lubricated and require no further lubrication. Brush SAE 40 oil on drive chain and sprocket chain every two weeks. Check level of oil in reduction drive housing once a month. Keep oil level to top of opening but no higher. Use No. 650 transmission oil. Drain once a year and refill. Grease the 14 grease fittings weekly. Fittings are located four at right side, nine at rear and one under right end of cylinder. Use a squirt type oil can with SAE 20 oil once a week at knife link, duster box levers, and conveyor pulleys.

3.7.3.3 Moulder.- The following instructions shall be incorporated on the plate shown on Drawing 5-11-845:

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OPERATING INSTRUCTIONS

Before starting moulder make sure duster box is free. Keep hands away from rolls while moulder is in operation. Sheet thickness is controlled by turning hand wheels on side of moulder. Wheels may be turned in either direction and thickness must be gaged by observation. Pressure board is adjusted by turning knobs at each end of moulder near bottom. Dusting flour should not be allowed to collect at pressure board. To shut off duster, pull out knob on duster level so that key is disengaged. Belt is adjusted by means of adjusting screws at idler-pulley end of moulder.

LUBRICATION INSTRUCTIONS

Motor ball-bearings are prepared and lubricated and require no further lubrication. Brush a small amount of SAE 40 oil on motor drive chain and sprocket chain every two weeks. Once a week, use a good grade of pressure gun grease on the 16 grease fittings located as follows: Two at each end of the moulder for the belt pulley; four on each side for the sheeting rolls; one behind motor for eccentric; one on each side near discharge for drive shaft; one through cover for idler. Use a squirt-type oil can with SAE 20 oil at duster level once a week.

3.8 Lighting.- Lighting on the machinery trailer shall conform to the design and materials specified on Drawing 5-11-855.

3.9 Performance.- When tested as specified in 4.4.3, the completely assembled outfit shall properly mix the ingredients and divide and mould the dough pieces.

3.10 Road and terrain.- The dough mixing and make-up outfit trailer mounted shall show no evidence of broken, missing or loose components when subjected to the road and terrain test specified in 4.4.1.

3.11 Registration numbers.- The registration numbers shall be furnished by the contracting officer (see 6.2) and shall be painted in 3-inch block letters and figures onto the chassis in three places as shown on Drawing 5-11-543, using white, lusterless, synthetic, stenciling enamel. For Marine Corps procurement the location of lettering and numerals shall be in accordance with MIL-STD-642.

3.12 Replacement part.- The following replacement motor shall be supplied with each outfit and shall be mounted in the repair parts box in the location provided as shown on Drawing 5-11-624.

Motor, 1-1/2 H.P., 3-phase, 208 volt, 60 hertz.

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* 3.13 Government furnished property.- The following property shall be furnished by the Government.

(a) Trailer flat bed, dough mixing and make-up outfit, 4-ton, 4-wheel, M-795 - Federal Stock Number (FSN) 2330-089-4321
(b) Compressor, reciprocating - FSN 4310-631-5693
(c) Gun, air blow - FSN 4940-241-3075
(d) Hose assembly 50 feet long, complete with 1/4-inch fittings - FSN 7360-300-0001
(e) Plug - FSN 5935-359-5990
(f) Power cable, 50 foot, 4 conductors with connectors - FSN 6150-767-7301

3.14 Workmanship.- The finished dough mixing and make-up outfit shall have no sharp burrs, edges or corners and shall have no slivers or splinters.

3.14.1 Welding and brazing.- All surfaces of parts welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.14.2 Soldering.- Soldering shall be smooth, sound and clean without pin holes.

3.14.3 Threaded fasteners.- Bolts, nuts, screws, studs and rivets shall be drawn up tight, and shall not be broken, stripped or fractured and washers shall not be missing.

3.14.4 Electrical assembly.- Wiring shall be properly joined, not be cut or abraded, not be loose at terminal end, be properly enclosed in conduit when required, be provided with slack, and not be stripped of required insulation.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 First article inspection.- When a preproduction sample is required, it shall be a completely assembled end item. The preproduction sample shall be examined for defects listed in table I, dimensions specified and tested as specified in 4.4.1, 4.4.2, 4.4.3 and 4.4.5. The presence of any defects listed in table I, any dimensions not as specified, or failure to pass any tests shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be made in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of reference specifications, drawings and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Chemical analysis.- The corrosion-resistant castings used for the mixer agitator elbow, sides, caps, moulder rollers and bowl rib shall be tested for chemical analysis in accordance with the procedure outlined in method 111.2 of FED-STD-151 to determine compliance with the requirements specified in 3.4.6.6. Any noncompliance shall be cause for rejection.

4.3.2 Process inspection.-

4.3.2.1 In-process examination.- Examination shall be made of the following operations to establish that no deviation is made from the specified requirements. Whenever a deviation is noted correction shall be made.

(a) Cleaning of surfaces of parts to be welded as specified in 3.5.12 and 3.14.1.

(b) Cleaning of surfaces of parts to be soldered as specified in 3.5.13.

(c) Welding as specified in 3.5.12.

(d) Cleaning, priming, and painting as specified in 3.6, 3.6.1, 3.6.2 and 3.6.3.

4.3.2.2 In-process testing.- In-process testing for hardness shall be as specified in 4.4.4. The sample unit shall be one roller, drive pinion, motor pinion or eccentric shaft, as applicable. The lot shall be the number of rollers, drive pinions, motor pinions or eccentric shafts offered for inspection at one time. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units.

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4.3.3 Intermediate testing.- The wood box and weight box shall be tested for moisture content in accordance with ASTM D 2016-65 for compliance with 3.4.13. The sample unit shall be one wood box or weight box, as applicable, and the lot shall be the number of wood boxes or weight boxes offered for inspection at one time. The test on the box shall be made prior to applying the primer coating. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.4 End product inspection.- The inspection lot shall consist of all dough mixing and make-up outfits offered for inspection at one time. The sample unit shall be one completely assembled end item.

4.3.4.1 Visual examination.- Examination shall be made of the end item for defects in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Type of finish not as specified.	X	
	Runs, wrinkles, chipping, or area of no film on enamel.	X	
	Not finished where required.	X	
	Component or assembly that is painted, or otherwise coated e.g., paint on gasket.		X
Construction and workmanship	Sharp burrs, slivers, splinters, edges, or corners.	X	
	Cotton and duck not as specified.	X	
Divider Conveyor belt	Conveyor belt not mounted below the dividing head.	X	
Moulder Rollers	Rollers not mounted with independent adjustment.	X	
	Scrapers missing.	X	
	Scrapers not mounted correctly		X

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Monorail			
Carrier rail	Locks, not provided to control movement of tramway.	X	
	Latches missing.	X	
Panel			
Panelboard	Panelboard not a dead front safety type.	X	
Cabinet	Safety door not provided.	X	
	Latches not positive in action.		X
Dusters	Dusters missing on divider and moulder.	X	
Gages and instruments	Not specified, capacity, or type, inoperative or defective.	X	
	Not mounted in specified manner.	X	
Covers, canvas	Canvas covers missing on mixer, divider and moulder or hopper	X	
Lighting kit	Missing.	X	
Welding and brazing (when applicable)	Missing, incomplete, burn holes, cracked, fractured, or not fused.	X	
	Slag inclusion, undercut, not smooth or uniform, scale, or flux deposits not removed	X	
Riveting	Rivet not peened.	X	
	Rivet peened improperly, or head cracked.	X	
	Rivet loose.		X
Lubrication (when applicable)	Lubrication fittings not readily accessible for cleaning.		X
	Lubrication fittings missing	X	
	Not lubricated.	X	
	Not specified type lubrication fittings.	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Soldering (when applicable)	Missing, not adherent, or incomplete.	X	
	Not clean, (flux or flux residue not removed), not smooth (surface not finished neatly) or pin holes in solder.	X	
Bolts, nuts, screws, washers, and other types of fasteners	Missing, broken, stripped or fractured.		X
	Loose.		X
	Washer missing.		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal, or not enclosed in specified conduit where required.	X	
	Adequate slack not provided for wiring to relieve strain or required insulation stripped from wiring.		X
Marking for identification and instruction plates	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner; plate not of specified material.		X
Instruction plates	Missing, incomplete, not as specified.	X	
USA Registration number	Incorrect, or not affixed as prescribed.	X	

4.3.4.2 Dimensional and coating thickness.- Examination shall be made for defects in dimensions and coating thickness. Any dimension or coating thickness that is not within the established requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.4.3 Testing of the end product.- Each end item shall be tested in accordance with 4.4.2. Any nonconformance shall be cause for rejection of the unit. When no preproduction sample is required, the initial preproduction unit shall be tested in accordance with 4.4.1, 4.4.3, and 4.4.5. Any nonconformance shall be cause for rejection of the unit.

* 4.3.5 Code and standard compliance.- Proof of compliance with the requirements of 3.3 and 3.4.6.6 shall be submitted to the Government representative.

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4.3.6 Examination of preparation for delivery.- Examination shall be made to determine that preservation and packaging are in accordance with MIL-P-116 and section 5. Examination for packing and marking shall be in accordance with PPP-C-650 and section 5.

4.4 Tests.-

4.4.1 15-Mile road and terrain.- The dough mixing and make-up outfit shall be towed to determine compliance with the requirements specified in 3.10, over not less than 5 miles of paved roads at speeds of 30 plus or minus 5 miles per hour (m.p.h.), as road and weather conditions permit; over not less than 5 miles of graveled roads at speeds averaging 20 m.p.h.; and over not less than 5 miles of open terrain at speeds averaging 5 m.p.h.

4.4.1.1 Paved and gravel roads.- The paved and gravel road test shall include not less than 20 quick starts and stops on grades of not less than 5 percent or greater than 10 percent. Fifty percent of the starts and stops shall be made on ascending and 50 percent on descending. Inability for the dough mixing and make-up outfit to withstand travel over paved and gravel roads shall constitute failure of this test.

4.4.1.2 Open terrain.- The open terrain shall have surface irregularities such as pot holes, mounds, rocks and brush randomly spaced. The test drive shall include 10 abrupt stops; 10 abrupt starts; and 10 sharp left hand turns and 10 sharp right hand turns. Inability of the dough mixing and make-up outfit to withstand travel over open terrain shall constitute failure of this test.

* 4.4.2 Operational test The complete dough mixing and make-up outfit shall be connected to power and operated without load to determine that all operating components and systems are in operable condition to determine compliance with 3.5 and 3.7.2.3. The operation test shall include but is not limited to the following: starting and stopping of each motor; turning all lights on and off; determination of working order of switches and controls; and determination of the motor current. Inability of the dough-mixing and make-up outfit to conform to the requirements of 3.5 and 3.7.2.3 shall constitute failure of the test.

4.4.3 Performance test.- The dough mixing and make-up outfit shall be connected to power and operated at capacity, mixing, fermenting, dividing, and moulding at least one full load of approximately 500 pounds of dough based on the straight field bread dough formula as detailed in Technical Manual TM-10-410, and shall be checked to determine conformance with 3.9. Necessary ingredients will be furnished by the supplier. The supplier shall furnish qualified bakery technicians for conducting this test. Inability of the dough-mixing and make-up outfit to conform to the requirements of 3.9 shall constitute failure of the test.

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4.4.4 Hardness.- The hardness of the roller, drive pinion, motor pinion and eccentric shaft shall be determined in accordance with ASTM Standard E18. One indentation shall be made on each end surface and one indentation in the middle surface to determine compliance with the requirement of 3.5.5.2 for the roller to have a hardness of 62-65 on the Rockwell C scale and to determine compliance with the requirements of 3.5.15 for the drive pinion, motor pinion and eccentric shaft to have a hardness of 52-60 on the Rockwell C scale. Inability to conform to the hardness requirements shall constitute failure of the test.

4.4.5 Electromagnetic compatibility.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the Contracting Office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Disassembly.- The divider-conveyor belt, moulder conveyor belt, curling screen and dough trough carriers shall be removed.

5.1.1.2 Preservation.- All preservation shall be in accordance with the material and method requirements of MIL-P-116 as hereinafter specified.

5.1.1.2.1 Cleaning.- All exposed uncoated ferrous metal surfaces of the trailer, equipment, and the uncoated ferrous metal surfaces accessible by doors, panels, lids, caps, covers, and plates, shall be cleaned process C-1 and thoroughly dried. This includes type 430 corrosion-resisting steel.

5.1.1.2.2 Application of preservative.- All cleaned surfaces normally coming in contact with the dough or its ingredients in operation, including bowls, agitators, hoppers, plungers, knives, rolls, pistons, reservoirs, cage assembles and curling screen shall be coated with type P-14 preservative. Cleaned surfaces that do not contact other surfaces or the dough in operation shall be coated with type P-1 perservative. The unpainted portions of the monorail shall be coated with P-1 preservative. Cleaned surfaces that contact other surfaces and not in dough in operation shall be coated with type P-6 preservative.

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5.1.1.2.3 Protection of preservative.- Surfaces from which preservative compounds might be removed by contact with packaging and packing materials shall be covered or wrapped with greaseproof barrier-material conforming to type I or II, grade A of MIL-B-121.

5.1.1.2.4 Sealing.- Lamps, electric cable connections, receptacles, switches, terminals, openings into motors, instruction and identification plates, control boxes, dials, gages, meters, locks, key openings, and crevices formed by doors, covers, or lids shall be completely sealed with tape conforming to PPP-T-60. Remaining openings into the units shall also be sealed with the same type tape as specified herein. All taped parts and all exposed wiring including covered cables shall be coated with varnish conforming to MIL-V-13811.

5.1.1.3 Lubrication.- Points on the chassis and the equipment, requiring lubrication shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.4 Packaging.-

5.1.1.4.1 Caution tag.- A red caution tag, giving explicit depreservation instructions shall be attached to each of the units on the trailer. These instructions shall included a warning to the effect that tape and all preservative compounds must be thoroughly removed from the entire unit and preserved parts, prior to starting and operating the equipment.

* 5.1.1.4.2 Monorail.- The monorail system shall be closed and locked in position. The support beams shall be strapped together in the locked position with four flat steel straps 3/4 by .0280 inch conforming to type I, grade B of QQ-S-781. The straps shall be located, one each, adjacent to each side of the wheel guide tracks. Cushioning material, fabricated from weather-resistant grade fiberboard or any equivalent material, shall be applied under straps at edges.

5.1.1.4.3 Leveling jacks.- The four leveling jacks shall be folded up and locked in position by means of the retainer lug and lock screws provided. The jack bases shall be wired to the jack shanks with not less than 0.072 inch diameter annealed wire to prevent movement while in transit. The long and short jack braces shall be secured in the brace retainer.

5.1.1.4.4 Hardware and scale.- All loose and dangling catches, holding hooks, safety chains, and hardware shall be wired in place to prevent movement while in transit. The scale table shall be depressed to rest on its stops and securely tied with cotton, jute, sisal, or manila twine or cord. The maintenance manual for the scales shall be packaged method IC-1 of MIL-P-116. The weights shall be immobilized within the weight box with a fiberboard pad. The volume of fluid required for each scale shall be filled into

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a leakproof container having a removable top for pouring purposes. Each glass fluid container shall be completely wrapped in minimum 1-inch thick cushioning material conforming to type I, class B of PPP-C-843 and then packed in a snug-fitting paperboard box conforming to any type, class, or style of PPP-B-566 or PPP-B-676. Closure of the box shall be in accordance with the appendix of the applicable box specification.

5.1.1.4.5 Curling screen and conveyor belting.- The preserved curling screen, the divider-conveyor belt and the moulder-conveyor belt shall each be rolled into a compact bundle and individually wrapped in greaseproof barrier-material conforming to type I or II, grade A, of MIL-B-121, and sealed with gummed paper tape conforming to type III, grade B of PPP-T-45. The curling screen, the divider-conveyor belt and the moulder-conveyor belt after wrapping, shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF variety optional, or type SF, class domestic, grade 200 of PPP-B-636. The box shall be sealed with gummed paper tape as specified herein and then completely wrapped with waterproof barrier material conforming to class L-2 or M-1 of PPP-B-1055. All seams and closures shall be completely sealed with water-resistant adhesive conforming to MMM-A-260. The packaged curling screen, divider-conveyor belt and moulder-conveyor belt shall be placed on the moulder-conveyor frame between the moulder and the curling screen supports, and held firmly in place with not less than 60-pound test jute twine to prevent movement while in transit.

5.1.1.4.6 Pressure board.- The pressure board shall be securely held in place so as to prevent movement while in transit.

5.1.1.4.7 Repair parts.- The technical manual packaged IC-1 shall be placed in the repair parts box. The replacement electric motor and the box thermometers shall be placed in their respective positions within the repair parts box. The repair parts box shall then be locked with the padlock provided for that purpose, the two keys wired to the lock with not less than 0.035 inch diameter annealed wire. The keys shall be covered with tape.

5.1.1.4.8 Dough trough carriers.- The five dough trough carriers, each with its four chains wound around the carrier beam and securely wired in place, shall be packed in a nailed wood crate conforming to type II of PPP-C-650. The five carriers shall be positioned in the crate, side by side, right side up. The second and fourth carrier shall be staggered several inches, resulting in a pack of minimum width. Blocking and bracing shall not depend on end grain nailing alone. Supplemental supporting cleats shall be provided to reinforce the blocks and prevent splitting of the blocking and bracing. The crated carriers shall be positioned flat wise across the trailer front, adjacent to and in front of the two forward monorail supports posts. Blocking and bracing shall

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be used in not less than two places on the trailer deck and on each of the front plates of the trailer frame so that the crate will clear the front wheel when it is in the raised position. The crate shall then be securely strapped to the trailer by passing metal straps the same size as used on the crate, around the end members of the crate and through the two front side lifting eyes. The crate shall be further secured by strapping to the two front monorail support posts and the front members of the trailer frame.

5.1.1.4.9 Rear platform.- The rear platform shall be raised and held securely in position by means of the retaining bolts provided for that purpose.

5.1.1.4.10 Lighting kit.- Each kit, packaged as specified on Drawing 5-13-2176, shall be further packaged in a snug-fitting fiberboard box conforming to style RSC, V3c of PPP-B-636. Each fiberboard box shall be closed in accordance with class weather-resistant box requirements in the appendix of PPP-B-636. The packaged kit shall be secured in an unused portion of trailer base.

5.1.2 Level C.- The trailer mounted dough mixing and make-up outfit shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A.

5.2.1 Level A.-

5.2.1.1 Covers.- The covers provided (see 3.4.25), shall be placed in position over the units and fastened in place with the billets going through the footman loops. The covers shall be further secured with not less than 60-pound test jute twine, which shall be passed over the covers in two directions and around the base. The twine shall be drawn tightly and tied through the billet loops.

5.2.1.2 Railway shipment.- For railway shipment, tire pressure shall be increased 10 pounds over the required running pressure. Loading of vehicles for railway shipment shall be on open-top cars in accordance with rules governing the loading of Department of Defense Materials on Open-Top Cars, and Pamphlet MD7 of the Association of American Railroads.

5.2.1.3 Highway haulaway shipment.- The units shall be placed on bed or rails of hauling vehicles so that the load is evenly distributed and secured. Parking brakes shall be locked and the levers wired or blocked in position. Loading of units for haulaway shall be in accordance with the Interstate Commerce Commission publication "Motors Carrier Safety Regulations". Tire pressure shall be increased 10 pounds over the required running pressure.

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5.2.1.4 Highway towaway shipment.- No processing of the trailer units running gear shall be performed prior to highway shipment that would impair necessary operation or safety during transit. Towaway rules for shipment by driveaway shall be in accordance with the Interstate Commerce Commission publication "Motor Carrier Safety Regulations". Tire pressure shall be the required running pressure. Processing of the trailer units running gear, not accomplished prior to highway shipment, shall be performed at intermediate destination when vehicles are to be placed in storage or shipped overseas.

5.3 Marking.- In addition to caution and deprocessing marking, interior packages and trailer mounted units shall be marked in accordance with the applicable requirements of MIL-STD-129. Special requirements applicable to unpacked vehicles shall apply.

5.3.1 Humping precaution.- A large sign containing the words "DO NOT HUMP" shall be secured in a conspicuous place on each side of open-top cars.

6. NOTES

* 6.1 Intended use.- The dough mixing and make-up outfit is intended for use as a component part of the mobile bakery unit. It also serves as a distribution station for power and lighting for the entire mobile bakery. The electrical power and illumination is provided from the electric generator specified in MIL-G-10327. A lighting kit for the illumination of the working area is also included (see 3.5.11). Five (5) dough troughs conforming to MIL-D-2222 are required for operation of the unit.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.2, 4.2 and 6.7).
(c) Registration number to be furnished contractor (see 3.11).
(d) Source of Government furnished property (see 3.13).
(e) When procurement is for Marine Corps (see 3.4.29.2).
(f) Selection of applicable levels of preservation and packaging (see 5.1).

6.3 Color chips for Marine Corps Green No. 23 may be obtained from the Commandant of the Marine Corps (Code CSY), Headquarters, Marine Corps, Washington, D. C., 20380.

6.5 Rubber.- White neoprene rubber CR-60W manufactured by the General Asbestos Gasket Manufacturing Corporation, 1761 South 7th Street, St. Louis, Missouri, 63104 has been found acceptable by the Surgeon General.

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6.6 Non-toxic grease.- Any question regarding the nontoxic greases shall be submitted to the Office of the Surgeon General by the procuring activity.

* 6.7 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of Armed Services Procurement Regulations 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.8 Contract data requirements.- Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

Custodian:

Army - GL
Navy - MC
Air Force - 84

Review activity:

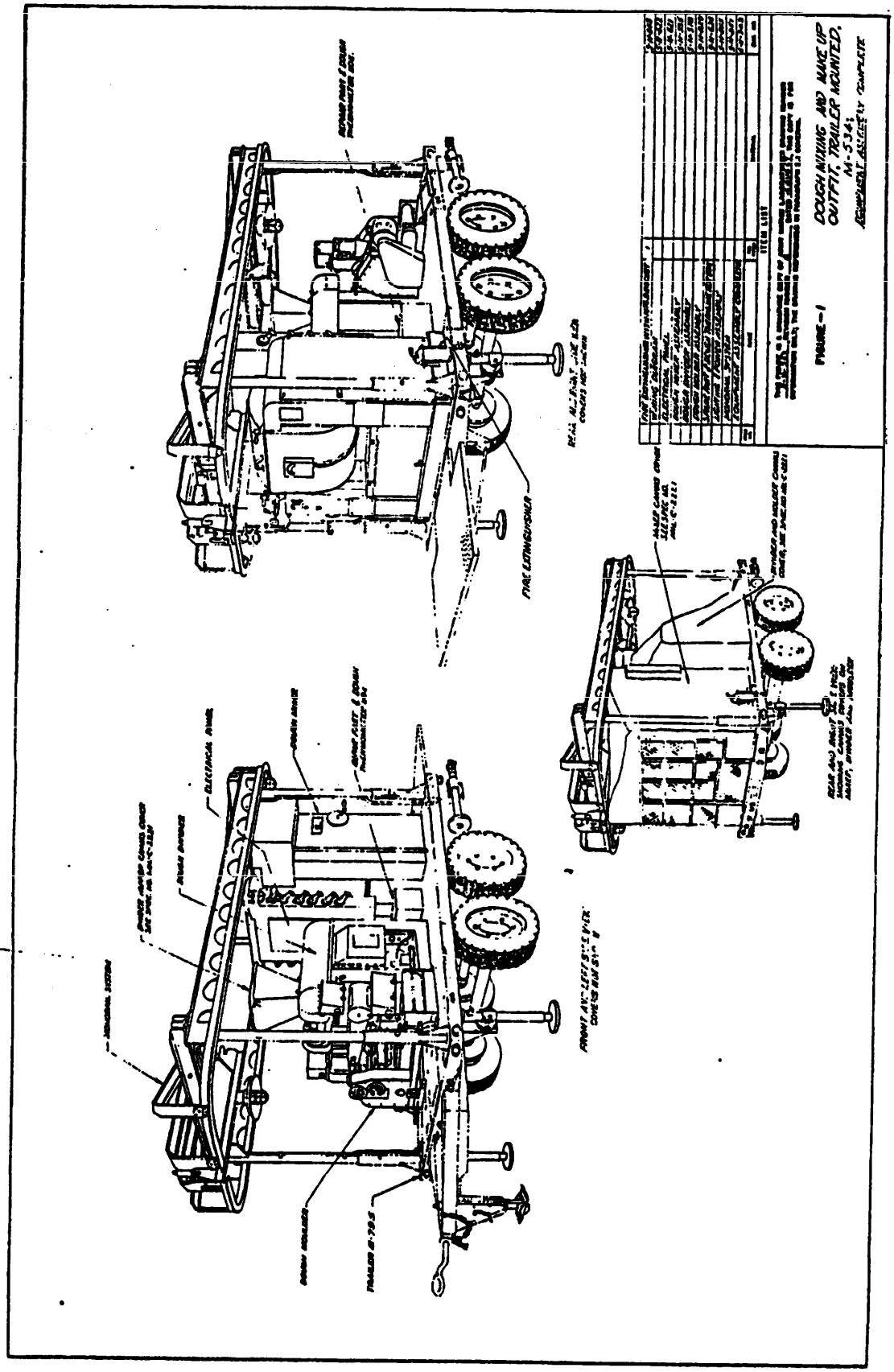
Army - MD

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Dough Mixing And Make-Up Outfit, Trailer-Mounted, M-534 MIL-D-43490A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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